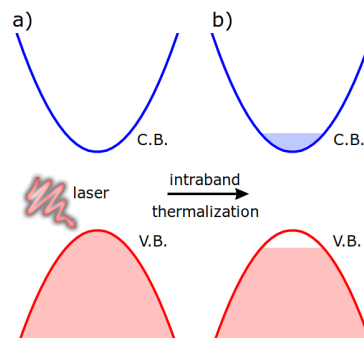


# Constrained Density Functional Perturbation Theory – Hands on

Giovanni Marini and Matteo Calandra



Giovanni Marini and Matteo Calandra, Phys. Rev. B **104**, 144103 (2021)

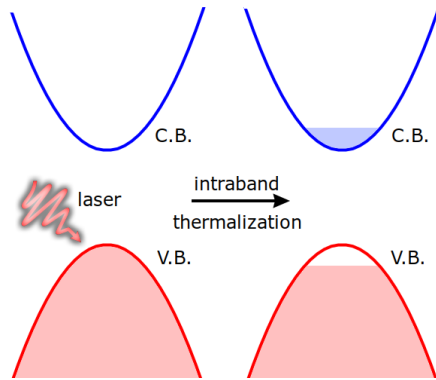


# How to efficiently simulate phonon and structural dynamics at very long times ?

## Constrained Density functional perturbation theory for photexcited insulators

Giovanni Marini and Matteo Calandra, Phys. Rev. B **104**, 144103 (2021)

- Complete cDFPT framework : two different chemical potentials and electron-hole plasma distribution, no supercells.



We build on top of Refs.:

S. De Gironcoli, Phys. Rev. B **51**, 6773(R) (1995)

Tangney & Fahy, PRL **82**, 4340 (1999), PRB Phys. Rev. B **65**, 054302 (2002)

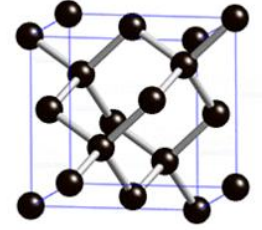


- Complete structural optimization (forces/cell optimization).
  - Vibrational properties (phonon dispersion)
  - Electron-phonon interaction (including photocarriers-photocarriers screening)
  - Coupling with SSCHA, Machine-learning based MD, *ab initio* BOMD, CPMD
- A. Corradini, G. Marini and M.C., npj Computational Materials **11**, 151 (2025)

How does all this framework works ?

# Structural Optimization – Example: Ground state Silicon

Input file PW\_Si\_GS/vc-relax.in



```
&control
  calculation = 'vc-relax'
  tprnfor=.true.
  tstress=.true.
  pseudo_dir='../Pseudo/'
/
&system
  ibrav=2, celldm(1)=10.20,
  nat=2, ntyp=1,
  ecutwfc=30.0
  occupations="fixed",
/
&electrons
/
&ions
/
&cell
/
ATOMIC_SPECIES
Si 28.086 Si.pz-vbc.UPF
ATOMIC_POSITIONS {alat}
Si 0.00 0.00 0.00
Si 0.25 0.25 0.25
K_POINTS automatic
4 4 4 1 1 1
```

→ Fixed occupation as the ground state is insulating.

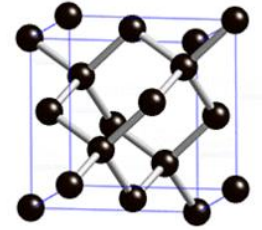
→ Perdew Zunger xc (LDA)

# Structural Optimization – Example: Ground state Silicon

## Output PW\_Si\_GS/vc-relax.out

```
bravais-lattice index   =      2
lattice parameter (alat) = 10.2000 a.u.
unit-cell volume       = 265.3020 (a.u.)^3
number of atoms/cell   =      2
number of atomic types  =      1
number of electrons     =     8.00
number of Kohn-Sham states =      4
```

....



This is the number of valence states.

As the system is insulating/semiconducting, no conduction states are needed.

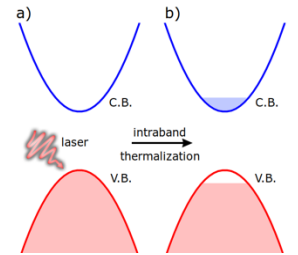
[Si]=[Ne]  $3s^2 3p^2$  has 4 electrons in valence, two atoms per cell=8 electrons,  
4 bands (two electrons per band).

# Structural Optimization – Example: Photoexcited Silicon

Input file PW\_Si\_Excited/vc-relax\_twochem.in

```
&control
  calculation = 'vc-relax'
  tprnfor=.true.
  tstress=.true.
  twochem=.true.
  pseudo_dir='./Pseudo/',
/
&system
 ibrav=2, celldm(1)=10.20,
 nat=2, ntyp=1,
 ecutwfc=30.0
 nbnd=8
 nbnd_cond=4
 nelec_cond=0.1
 occupations="smearing"
 degauss_cond=0.01
 degauss=0.01
 smearing='gaussian'
/
&electrons
/
&ions
/
&cell
/
ATOMIC_SPECIES
Si 28.086 Si.pz-vbc.UPF
ATOMIC_POSITIONS {alat}
Si 0.00 0.00 0.00
Si 0.25 0.25 0.25
K_POINTS automatic
4 4 4 1 1 1
```

Changes are concentrated in these two blocks



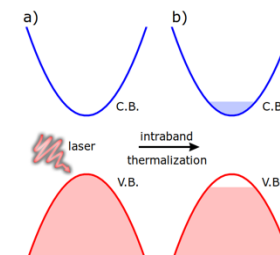
# Structural Optimization – Example: Photoexcited Silicon

Input file PW\_Si\_Excited/vc-relax\_twochem.in

```
&control
  calculation = 'vc-relax'
  tprnfor=.true.
  tstress=.true.
  twochem=.true.
  pseudo_dir='./Pseudo/',
/
&system
 ibrav=2, celldm(1) =10.20,
 nat=2, ntyp=1,
 ecutwfc=30.0
 nbnd=8
 nbnd_cond=4
 nelec_cond=0.1
 occupations="smearing"
 degauss_cond=0.01
 degauss=0.01
 smearing='gaussian'
/
&electrons
/
&ions
/
&cell
/
ATOMIC_SPECIES
Si 28.086 Si.pz-vbc.UPF
ATOMIC_POSITIONS {alat}
Si 0.00 0.00 0.00
Si 0.25 0.25 0.25
K_POINTS automatic
4 4 4 1 1 1
```



Flag activating the two chemical potentials option in pw.x



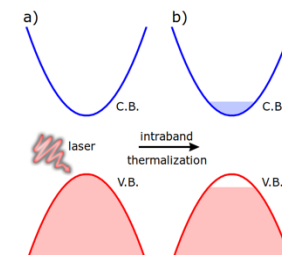
# Structural Optimization – Example: Photoexcited Silicon

Input file PW\_Si\_Excited/vc-relax\_twochem.in

```
&control
  calculation = 'vc-relax'
  tprnfor=.true.
  tstress=.true.
  twochem=.true.
  pseudo_dir='./Pseudo/',
/
&system
 ibrav=2, celldm(1)=10.20,
nat=2, ntyp=1,
ecutwfc=30.0
nbnd=8
nbnd_cond=4
nelec_cond=0.1
occupations="smearing"
degauss_cond=0.01
degauss=0.01
smearing='gaussian'
/
&electrons
/
&ions
/
&cell
/
ATOMIC_SPECIES
Si 28.086 Si.pz-vbc.UPF
ATOMIC_POSITIONS {alat}
Si 0.00 0.00 0.00
Si 0.25 0.25 0.25
K_POINTS automatic
4 4 4 1 1 1
```



We need to include more bands as we want to populate the conduction band

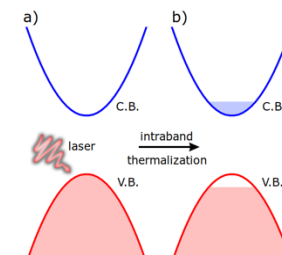


# Structural Optimization – Example: Photoexcited Silicon

Input file PW\_Si\_Excited/vc-relax\_twochem.in

```
&control
  calculation = 'vc-relax'
  tprnfor=.true.
  tstress=.true.
  twochem=.true.
  pseudo_dir='./Pseudo/',
/
&system
 ibrav=2, celldm(1) =10.20,
 nat=2, ntyp=1,
 ecutwfc=30.0
 nbnd=8
 nbnd_cond=4
 nelec_cond=0.1
 occupations="smearing"
 degauss_cond=0.01
 degauss=0.01
 smearing='gaussian'
/
&electrons
/
&ions
/
&cell
/
ATOMIC_SPECIES
Si 28.086 Si.pz-vbc.UPF
ATOMIC_POSITIONS {alat}
Si 0.00 0.00 0.00
Si 0.25 0.25 0.25
K_POINTS automatic
4 4 4 1 1 1
```

→ We need to tell pw.x how many of these bands belong to the conduction, this means The last four bands (5,6,7,8) belongs to the conduction.



# Structural Optimization – Example: Photoexcited Silicon

Input file PW\_Si\_Excited/vc-relax\_twochem.in

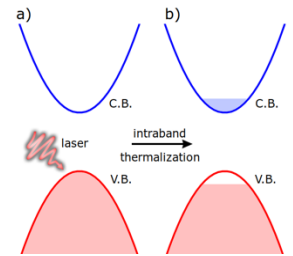
```
&control
  calculation = 'vc-relax'
  tprnfor=.true.
  tstress=.true.
  twochem=.true.
  pseudo_dir='./Pseudo/',
/
&system
 ibrav=2, celldm(1) =10.20,
 nat=2, ntyp=1,
 ecutwfc=30.0
 nbnd=8
 nbnd_cond=4
 nelec_cond=0.1
 occupations="smearing"
 degauss_cond=0.01
 degauss=0.01
 smearing='gaussian'
/
&electrons
/
&ions
/
&cell
/
ATOMIC_SPECIES
Si 28.086 Si.pz-vbc.UPF
ATOMIC_POSITIONS {alat}
Si 0.00 0.00 0.00
Si 0.25 0.25 0.25
K_POINTS automatic
4 4 4 1 1 1
```



As we transferred electrons in the conduction band, leaving holes in the valence, The system is metallic and we need to treat the system as a metal.

In principle we can use different values for the electrons broadening (degauss\_cond) And for the holes broadening (degauss)

In the present case they are identical.



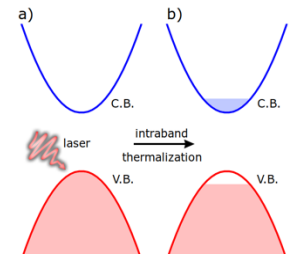
# Structural Optimization – Example: Photoexcited Silicon

Input file PW\_Si\_Excited/vc-relax\_twochem.in

```
&control
  calculation = 'vc-relax'
  tprnfor=.true.
  tstress=.true.
  twochem=.true.
  pseudo_dir='./Pseudo/',
/
&system
 ibrav=2, celldm(1)=10.20,
nat=2, ntyp=1,
ecutwfc=30.0
nbnd=8
nbnd_cond=4
nelec_cond=0.1
occupations="smearing"
degauss_cond=0.01
degauss=0.01
smearing='gaussian'
/
&electrons
/
&ions
/
&cell
/
ATOMIC_SPECIES
Si 28.086 Si.pz-vbc.UPF
ATOMIC_POSITIONS {alat}
Si 0.00 0.00 0.00
Si 0.25 0.25 0.25
K_POINTS automatic
4 4 4 1 1 1
```

Number of electron per cell in the conduction band.

For charge conservation, the same number of holes will be put in the valence band.

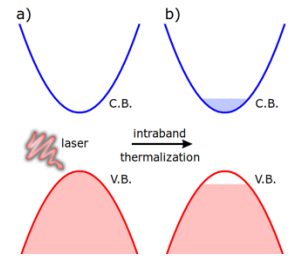


# Structural Optimization – Example: Photoexcited Silicon

Output file PW\_Si\_Excited/vc-relax\_twochem.out

bravais-lattice index = 2  
lattice parameter (alat) = 10.2000 a.u.  
unit-cell volume = 265.3020 (a.u.)<sup>3</sup>  
number of atoms/cell = 2  
number of atomic types = 1  
number of electrons = 8.00  
number of Kohn-Sham states = 8

→ Increased number of KS states including conduction states

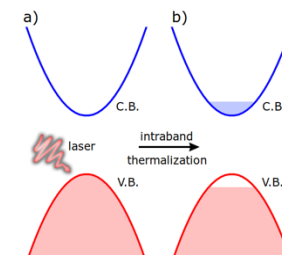


# Structural Optimization – Example: Photoexcited Silicon

Output file PW\_Si\_Excited/vc-relax\_twochem.out

bravais-lattice index = 2  
lattice parameter (alat) = 10.2000 a.u.  
unit-cell volume = 265.3020 (a.u.)<sup>3</sup>  
number of atoms/cell = 2  
number of atomic types = 1  
number of electrons = 8.00  
number of Kohn-Sham states = 8

→ Increased number of KS states including conduction states



-----2CHEM-----

You are performing a constrained density-functional perturbation theory  
employing two chemical potentials, one for electrons and one for holes.

Please refer to:

Giovanni Marini, Matteo Calandra

Lattice dynamics of photoexcited insulators

constrained density-functional perturbation theory

Phys. Rev. B 104, 144103 (2021)

doi:10.1103/PhysRevB.104.144103

The conduction manifold is constituted by 4 bands  
0.1000 electrons are placed in the conduction manifold

-----2CHEM-----

→ Number of conduction band states and  
→ Number of photoexcited electrons

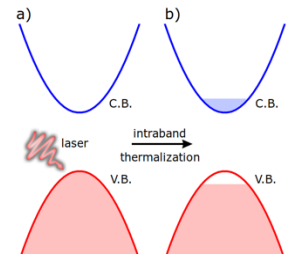
# Structural Optimization – Example: Photoexcited Silicon

Output file PW\_Si\_Excited/vc-relax\_twochem.out

## First iteration

Computing stress (Cartesian axis) and pressure

| total stress (Ry/bohr**3) |             |             | (kbar) | P=    | -4.26 |
|---------------------------|-------------|-------------|--------|-------|-------|
| -0.00002894               | 0.00000000  | 0.00000000  | -4.26  | 0.00  | 0.00  |
| -0.00000000               | -0.00002894 | 0.00000000  | -0.00  | -4.26 | 0.00  |
| 0.00000000                | 0.00000000  | -0.00002894 | 0.00   | 0.00  | -4.26 |



Note the appearance of a non-zero stress at the first iteration due to the presence of photoexcited carriers.

# Structural Optimization – Example: Photoexcited Silicon

Output file PW\_Si\_Excited/vc-relax\_twochem.out

## First iteration

Computing stress (Cartesian axis) and pressure

| total stress (Ry/bohr**3) |             |             | (kbar) | P=    | -4.26 |
|---------------------------|-------------|-------------|--------|-------|-------|
| -0.00002894               | 0.00000000  | 0.00000000  | -4.26  | 0.00  | 0.00  |
| -0.00000000               | -0.00002894 | 0.00000000  | -0.00  | -4.26 | 0.00  |
| 0.00000000                | 0.00000000  | -0.00002894 | 0.00   | 0.00  | -4.26 |

## Final iteration

Computing stress (Cartesian axis) and pressure

| total stress (Ry/bohr**3) |             |             | (kbar) | P=    | -0.38 |
|---------------------------|-------------|-------------|--------|-------|-------|
| -0.00000259               | 0.00000000  | -0.00000000 | -0.38  | 0.00  | -0.00 |
| 0.00000000                | -0.00000259 | -0.00000000 | 0.00   | -0.38 | -0.00 |
| -0.00000000               | -0.00000000 | -0.00000259 | -0.00  | -0.00 | -0.38 |

## Converged structure

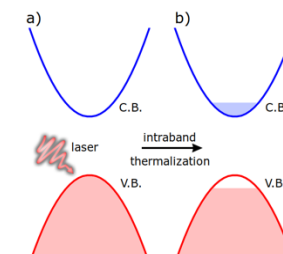
CELL\_PARAMETERS (alat= 10.20000000)

|              |             |              |
|--------------|-------------|--------------|
| -0.499316644 | 0.000000000 | 0.499316644  |
| 0.000000000  | 0.499316644 | 0.499316644  |
| -0.499316644 | 0.499316644 | -0.000000000 |

ATOMIC\_POSITIONS (alat)

|    |               |               |              |
|----|---------------|---------------|--------------|
| Si | -0.0000000000 | -0.0000000000 | 0.0000000000 |
| Si | 0.2496583221  | 0.2496583221  | 0.2496583221 |

End final coordinates



Structural minimization accomplished.

Initial (Ground state)  $a = 2.6988 \text{ \AA}$   
Converged (Exc. State)  $a = 2.695114 \text{ \AA}$

Si contracts when photoexcited.

# Structural Optimization – Homework

Try to reproduce the following results in

Giovanni Marini and Matteo Calandra, Phys. Rev. B **104**, 144103 (2021)

TABLE II. Structural parameters as a function of PC for tellurium, silicon, and gallium arsenide in the cDFPT approach.

| $n_e$ ( $e/\text{atom}$ ) | 0     | 0.015 | 0.03  | 0.045 | 0.06  |
|---------------------------|-------|-------|-------|-------|-------|
| $a$ (Te) (Å)              | 4.502 | 4.506 | 4.52  | 4.533 | 4.546 |
| $c/a$ (Te)                | 1.325 | 1.31  | 1.29  | 1.27  | 1.25  |
| $x$ (Te, crystal)         | 0.271 | 0.275 | 0.281 | 0.287 | 0.293 |
| $n_e$ ( $e/\text{atom}$ ) | 0     | 0.05  | 0.1   | 0.15  | 0.2   |
| $a$ (Si) (Å)              | 5.381 | 5.372 | 5.365 | 5.361 | 5.359 |
| $a$ (GaAs) (Å)            | 5.527 | 5.541 | 5.561 | 5.583 | 5.623 |

# Electronic structure, ground state v.s. photoexcited silicon

PW\_Si\_Bands\_Excited/

The electronic structure is obtained using the same procedure as usual  
But using the new flags we studied before.

```
mpirun -n 24 pw.x -npool 4 -in scf_twochem.in > scf_twochem.out  
mpirun -n 24 pw.x -npool 4 -in nscf_twochem.in > nscf_twochem.out  
mpirun -n 24 bands.x -npool 4 -in bands.in > bands.out
```

...

```
the Fermi energy is 5.8489 ev  
the conduction Fermi energy is 7.0813 ev
```

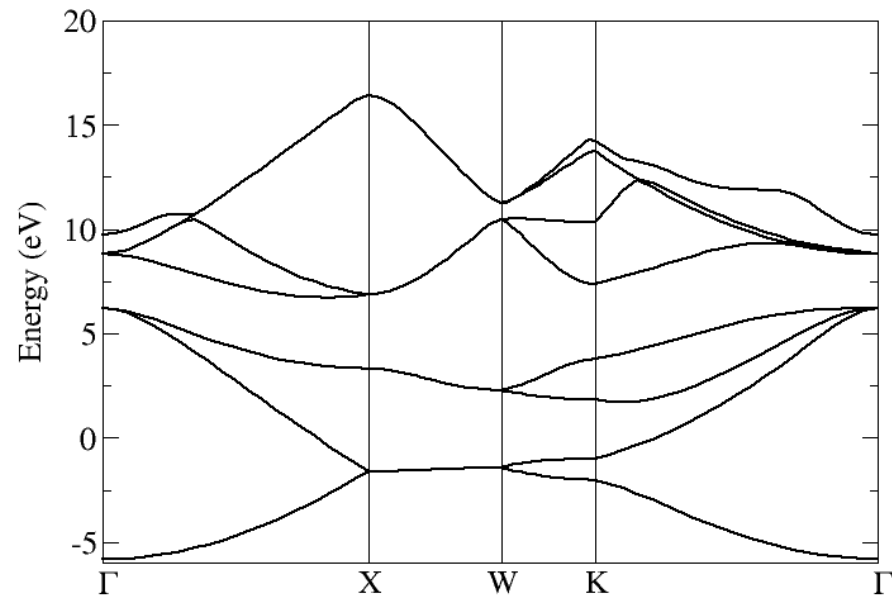
Valence and conduction band fermi levels.

```
! total energy = -15.84463702 Ry
```

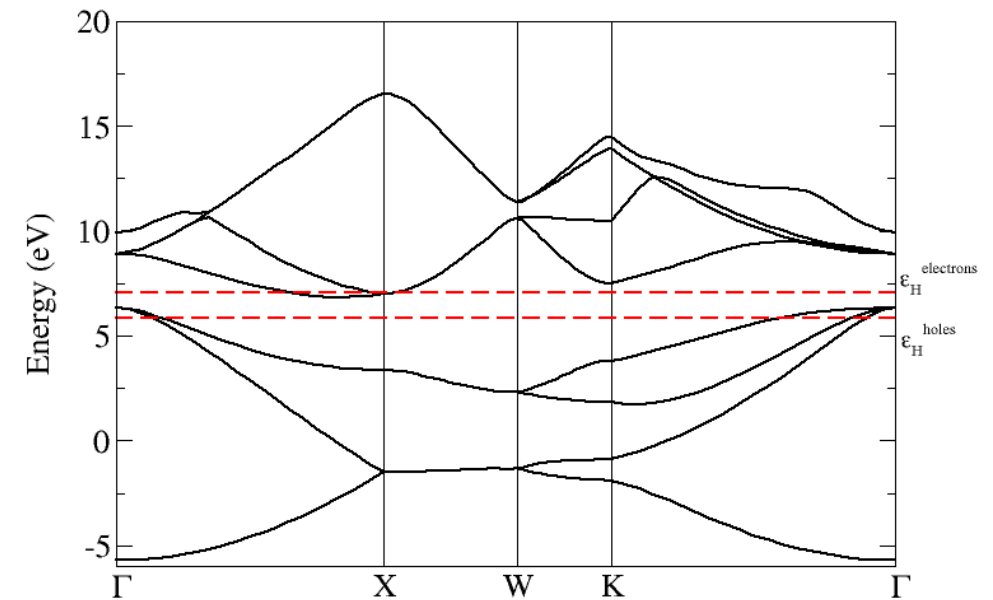
...

# Electronic structure, ground state v.s. photoexcited silicon

Ground state



Photexcited state



Note the very small holes and electrons Fermi surfaces in the photoexcited state!

Generally very large electron and phonon momenta grids are needed to converge photoexcited calculations!

Once the flags in the pw.x scf calculation has been set for the twochem option nothing is needed in the phonon calculation.

```
mpirun -n 24 pw.x -npool 4 -in scf_twochem.in > scf_twochem.out  
mpirun -n 24 ph.x -npool 4 -in ph_twochem.in > ph_twochem.out
```

The phonon input is the same as for the ground state (remember to converge k and q points) and all options valid for the ground state in ph.x are still valid for the excited state.

```
si  
&inputph  
  amass(1)= 28.086,  
  fildyn='dynq1',  
  fildvscf='dvscf',  
  outdir='./',  
  alpha_mix(1)=0.2  
  tr2_ph=1.0d-16,  
  trans=.true.,  
/  
0.0 0.0 0.0
```

## cDFPT – phonon calculation

### Ph.x output

Pert. # 1: Fermi energy shift valence (Ry) = 0.0000E+00 -2.4489E-39  
Pert. # 2: Fermi energy shift valence (Ry) = -2.0961E-18 2.5102E-39  
Pert. # 3: Fermi energy shift valence (Ry) = -2.8912E-18 -4.8979E-40  
Pert. # 1: Fermi energy shift conduction (Ry) = 1.8070E-20 0.0000E+00  
Pert. # 2: Fermi energy shift conduction (Ry) = -5.4210E-20 5.7397E-42  
Pert. # 3: Fermi energy shift conduction (Ry) = -3.6140E-20 -1.5306E-41

At  $q=0$  the Fermi shift occurs both for  
Valence and conduction states.

**WARNING** : in espresso 7.5 a bug was introduced for systems with non-zero fermi shifts.  
The bug has been corrected in version 7.5NC  
The version 7.4 is also correct

# cDFPT – phonon calculation

## Excited state

Diagonalizing the dynamical matrix

q = ( 0.000000000 0.000000000 0.000000000 )

```
*****
freq ( 1) = -0.303738 [THz] = -10.131598 [cm-1]
freq ( 2) = -0.303738 [THz] = -10.131598 [cm-1]
freq ( 3) = -0.303738 [THz] = -10.131598 [cm-1]
freq ( 4) = 15.285581 [THz] = 509.872114 [cm-1]
freq ( 5) = 15.285581 [THz] = 509.872114 [cm-1]
freq ( 6) = 15.285581 [THz] = 509.872114 [cm-1]
*****
```

## Excited state

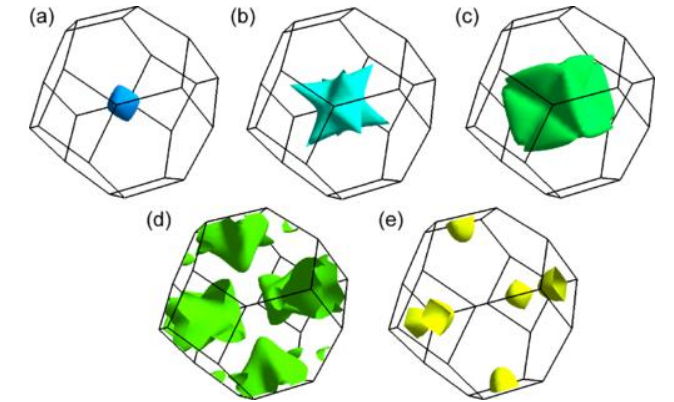
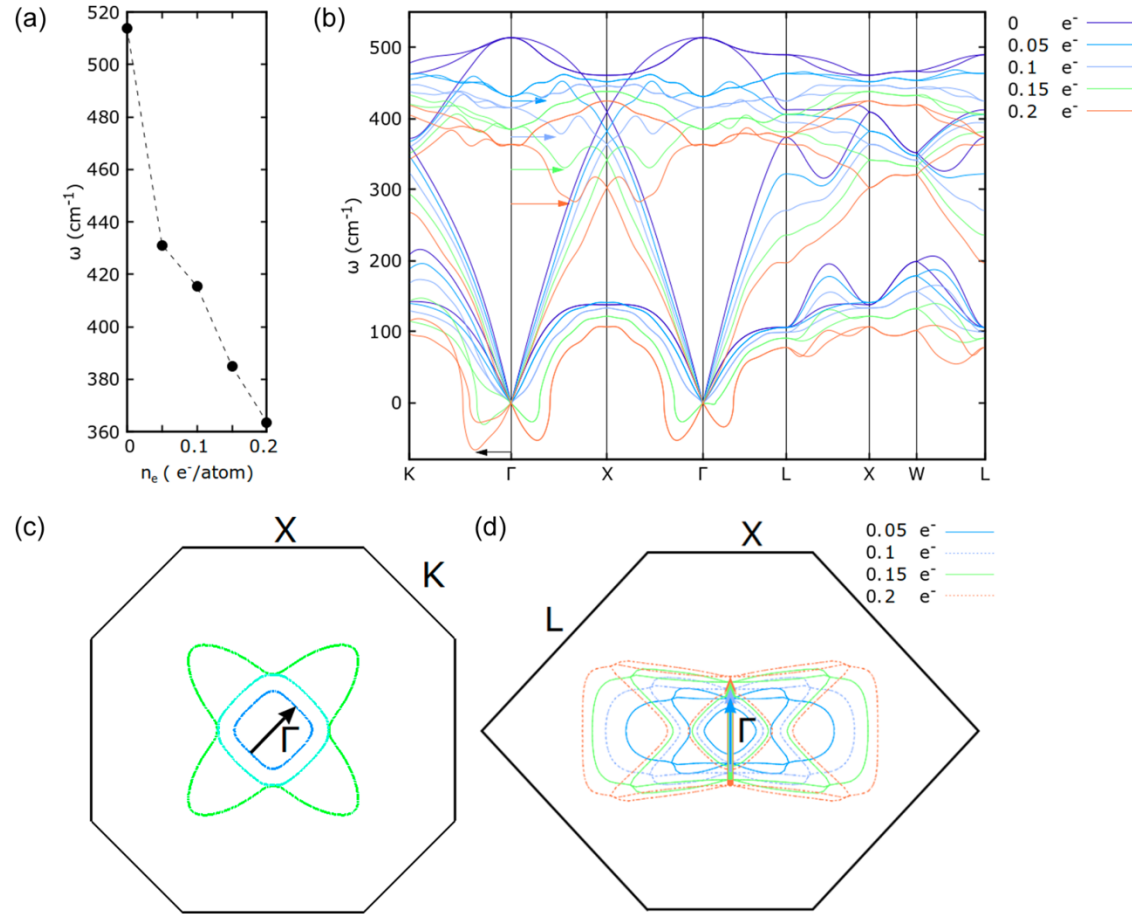
Diagonalizing the dynamical matrix

q = ( 0.000000000 0.000000000 0.000000000 )

```
*****
freq ( 1) = -0.436693 [THz] = -14.566520 [cm-1]
freq ( 2) = -0.436693 [THz] = -14.566520 [cm-1]
freq ( 3) = -0.436693 [THz] = -14.566520 [cm-1]
freq ( 4) = 12.556812 [THz] = 418.850152 [cm-1]
freq ( 5) = 12.556812 [THz] = 418.850152 [cm-1]
freq ( 6) = 12.556812 [THz] = 418.850152 [cm-1]
*****
```

Note the strong softening of the optical modes due to photoexcitation

# cDFPT – phonon calculation Si



(a)–(c) The three sheets of the valence quasi-FS for silicon at a PC value of 0.2  $e/\text{atom}$  (d) and (e) The two sheets of the conduction quasi-FS at a PC value of 0.2  $e/\text{atom}$  FS

FIG. 4. (a) Optical phonon frequency for silicon at  $\Gamma$  as a function of PC. (b) Phonon frequencies for silicon at selected PC values. (c) Section of the valence quasi-FS at  $n_e = 0.2$   $e/\text{atom}$ , in the plane perpendicular to the [110] reciprocal direction. The black arrow indicates the nesting vector responsible for the phonon instability. (d) Section of the valence quasi-FS at various PC values, in the plane perpendicular to the [211] reciprocal direction. Arrows indicate nesting vectors responsible for the softenings observed in the optical branch along the  $\Gamma$ -X direction. FS plots have been realized using the open-source software FERMISURFER [60].

## cDFPT – Conclusions

Remember to cite

Giovanni Marini and Matteo Calandra, Phys. Rev. B **104**, 144103 (2021)

And quantum espresso



When you use cDFPT be careful in what follows:

1. The code is intended to work for **systems that are gapped after photoexcitation**. If the gap closes after photexcitation (usually the gap is reduced by photoexcitation) it could work, depends on what happens exactly.
2. Be careful with convergence. Photoexcitation creates small Fermi surfaces and calculations requires **very dense electron and phonon momentum grids**.
3. Forces can be used to develop machine learning potential in the presence of photoexcitation see for example:

A. Corradini, G. Marini and M.C., npj Computational Materials 11, 151 (2025)